KSENIA SHPERLING

SOFTWARE DEVELOPER | PYTHON, AI INTEGRATIONS & SEMANTIC SEARCH

Moscow, Russia | +7 987 397 11 14 | Available for full-time remote work

PROFILE

Software Engineering student at HSE University, graduating in 2027, and Junior Analyst contributing to an AI assistant. Hands-on university project experience in embedding-based semantic search, LLM API integration, Python/FastAPI backend development, and React frontend development. Independently developed a visual novel framework in C# and Unity. Strong foundation in mathematics, algorithms, data processing, testing, and debugging.

TECHNICAL SKILLS

Programming: Python, C#, SQL, C++; JavaScript/TypeScript through React project work; additional familiarity with Swift.

AI & data: LLM API integration, OpenAI-compatible APIs (BotHub), embeddings, semantic search, PostgreSQL/pgvector, Pandas, NumPy, data cleaning and preprocessing.

Application development: FastAPI, REST APIs, JSON, React, PostgreSQL, MySQL, Docker, Git; software testing, code review, debugging, algorithms and data structures.

Game development: Unity, C#, Live2D integration, Pygame, Unreal Engine; basic Blender scene export.

EXPERIENCE

Junior Analyst | Rikor Holding

Oct 2023 - Present

- Contribute to Lara, an AI assistant, through software testing, code review, debugging, and bug fixing.
- Clean and preprocess data for analysis; translate tasks into algorithmic steps and use AI tools to help draft tests and test examples.

Teaching Assistant, Mathematical Analysis | HSE University

Sep 2024 - Present

- Support 30+ first-year students through consultations, homework grading, and written and oral assessments.
- Explain mathematical concepts and problem-solving approaches clearly; respond to student questions.

SELECTED PROJECTS

AI-Powered Job Platform | University Team Project

Python, FastAPI, React, TypeScript, PostgreSQL, pgvector

- Developed the frontend for a recruitment platform with AI-assisted resume analysis, job matching, and interview preparation features.
- Implemented embedding-based semantic search for jobs and resumes using PostgreSQL and pgvector.
- Contributed to backend development and LLM API integration through BotHub using the OpenAI SDK.
- Project repository: gitlab.com/matveipasynkov/job-platform-ai

Visual Novel Framework | Independent University Project

C#, Unity, Live2D

- Independently wrote the application code, including a dialogue script parser, extensible command system, and character management.
- Implemented animated text display, coroutine-based dialogue and command execution, character transitions, and Live2D expression and animation controls.

EDUCATION & LANGUAGES

HSE University, Faculty of Computer Science | Moscow, Russia
Bachelor's degree in Software Engineering | Expected graduation: 2027

Academic foundation: Mathematical analysis, linear algebra, discrete mathematics, probability, statistics, automata and process models, and economics.

Languages: Russian - Native | English - Upper-intermediate (B2)

ADDITIONAL EXPERIENCE

Personal game development since 2020 with Python/Pygame, Unity, and Unreal Engine. Independent blogging and content creation since 2018: research, copywriting, social media, audience engagement, photo/video content, and visual design. Additional experience in event organization and creative projects.